Post-Quantum Authentication and Communication Security for Drone Networks

Parya Derakhshan Roodsari, Siavash Bayat Sarmadi, and Hatame Mosanaei Boorani

***Abstract*—Modern societies increasingly rely on interconnected devices and individuals for information exchange and monitoring. In this manner, drones, as flying smart nodes in the wireless network ecosystem, facilitate diverse applications, such as industrial, civilian, and disaster response operations. However, the widespread adoption of drone networks introduces significant challenges regarding interoperability, privacy, and security. Advances in quantum computing exacerbate these issues by rendering classical cryptographic measures inadequate, necessitating the adoption of quantum-safe approaches. Existing solutions fail to address all potential threats or lack post-quantum security. This work sets out to propose a robust lattice-based authentication and communication protocol for drone networks, addressing these challenges. The protocol integrates dynamic credentials, timestamps, multi-factor authentication, and context information to enhance security. The scheme's security is analyzed using the DY and BPR models and formally verified via the AVISPA tool. Performance evaluations and comparisons with existing solutions demonstrate the protocol's functionality, showcasing a reasonable trade-off between security and performance. By leveraging modern cryptographic techniques, the proposed scheme ensures reliable and robust drone operations, effectively mitigating various attacks beyond session establishment while safeguarding privacy, and supporting dynamic network events.**

***Index Terms*—Drone networks, Internet of Drones, post-quantum cryptography, lattice-based cryptography, authentication, security.**

I. Introduction

DRONES or Unmanned Aerial Vehicles (UAV) are small devices capable of rapid deployment at high elevation angles. Owning to their capacity for dynamic reconfiguration, rapid response, and easy deployment, drones are considered a paramount solution in numerous domains, such as imagery, rescue, cargo delivery, agriculture, and surveillance [1]. Drones are equipped with sensors to collect data and measure physical attributes, such as altitude, speed, and location [2]. Drone networks are a subclass of Flying Ad-hoc Networks (FANET), which were designed to address the unique challenges and requirements of airborne platforms in a multi-hop ad-hoc manner. Internet of Drones (IoD) is an enhanced "layered network control architecture", playing a crucial role in UAV coordination [3].

Drones' high mobility, speed, node density and dynamic distribution, limited energy, computational power, broadcast vulnerability, and mission-critical security demands, differentiates them from other ad hoc devices in Internet of Things (IoT) and Wireless Sensor Networks (WSN) [4]. UAVs often operate in hostile or unsupervised environments, facing insecure wireless communication links and are susceptible to attacks due to their limited resources [5]. Attackers can use such weaknesses to access sensitive data or take control of critical infrastructure. Typical paradigms of attacks on drone networks are eavesdropping, tampering/spoofing, replay, man-in-the-middle (MITM), impersonation, tracking, drone capture, stolen verifier, ephemeral secret leakage, de-authentication, and login without verification attacks [6], [7], [8]. Due to the possible economic, social, and environmental ramifications of such attacks, mitigating potential threats is crucial.

In security solutions, confidentiality, integrity, and availability can be guaranteed through mutual authentication, forward security, secure key establishment, and anonymity or untraceability properties [2]. Establishing a shared secret securely over public channels is a crucial aspect of authentication and secure communication. This is typically achieved through authenticated key exchange (AKE) protocols based on public key cryptography [3]. Multi-factor authentication (MFA) further enhances security by incorporating independent components such as knowledge (e.g., password or ID), possession (e.g., certificate), or inherent (e.g., bio-metric) factors during authentication [1].

Classical cryptography schemes rely on the integer factorization, discrete logarithm, or elliptic-curve problems which can be broken in polynomial time by quantum computers. Post-quantum cryptography (PQC), also referred to as quantum-safe or quantum-resistant, is the development of algorithms assumed secure against quantum threats. The U.S. National Institute of Standards and Technology (NIST) has selected a key encapsulation mechanism (KEM) based on the Module Learning with Errors (MLWE) problem for PQC standards [9].

Motivations: Unlike most ad hoc networks such as WSN or IoT, drone communication lacks standardization and unification. The heterogeneous nature and diverse applications of drone networks, and challenges like resource limitations, high mobility, and untrusted dynamic environments, necessitate flexible and specialized security solutions. Robust node authentication and message integrity verification amidst extensive network events—i.e., registration, re-engagement, and revocation—are the prime security requirements. Existing solutions often lack quantum resistance, fail to address drone-specific characteristics, are not context-aware, lack alignment with standards, or do not mitigate all threats, rendering them insufficient.

Contributions: This work presents a standard-based, comprehensive protocol for post-quantum authentication and communication security in drone networks, with security prioritized over performance. Our key contributions include:

Received 6 January 2025; revised 26 August 2025; accepted 16 November 2025. Date of publication 25 November 2025; date of current version 12 March 2026. *(Corresponding author: Siavash Bayat Sarmadi.)*

The authors are with the Department of Computer Engineering, Sharif University of Technology, Tehran 14588-51147, Iran (e-mail: parya.derakhshan79@sharif.edu; sbayat@sharif.edu; h.mosanaei94@sharif.edu).

Digital Object Identifier 10.1109/TDSC.2025.3636937

1545-5971 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

- Designing a lattice-based AKE scheme aligned with NIST's quantum-safe standards.
- Addressing prior works' limitations, vulnerabilities, quantum insecurity, and lack of drone-specific design, while enhancing the existing PQC-based IoD scheme's functionality and security.
- Leveraging MLWE-based cryptography, key encapsulation, error reconciliation, and hash functions for post-quantum security.
- Integrating timestamps, pseudo-identities, and ephemeral credentials to mitigate various attacks.
- Enabling multi-factor authentication to enhance security.
- Using a fuzzy extractor for user's bio-metric authentication and secure storage on smart devices.
- Leveraging context information for periodic integrity verification and anomaly detection, tailored to drones' unique traits like mobility and mission-safety in hostile environments
- Securing dynamic drone/user operations.

The scheme is analyzed under the established BPR [10] and DY [11] models, and formally verified using AVISPA [12]. Comparative analysis confirms stronger quantum-security guarantees and improved functionality, with reasonable performance trade-offs.

Outline: The remaining sections are arranged as follows: Section II discusses an overview of related work; Section III introduces preliminaries; Section IV presents the network and security models; Section V describes the proposed scheme, procedures, and proof of correctness; Section VI analyzes security; Section VII evaluates and compares security, and performance; and Section VIII summarizes the contributions and outlines future work.

II. Related Work

Security measures in drone networks typically rely on hardware (e.g., information theory security, anti-spoofing schemes, or anti-jamming approaches) or software and cryptography [1]. Some of these solutions are discussed in this section.

The scheme proposed by [13] uses a cost-effective **Physical Unclonable Function** (PUF) eliminating the need for key storage. However, PUF's sensitivity to environmental changes can hinder proper key generation. [14] and [15]'s **data-type**-aware AKE protocols for IoD, use temporal credentials and PUFs, respectively, but they lack quantum security guarantees.

Blockchain-based approaches like [16] can provide robust security features, but increasing the number of block transactions increases the computation overhead. Several **machine learning**-based schemes, such as [17] and [18], have been suggested for the real-time identification of UAVs and intruders; however, to gain practical feasibility in such solutions, a significant amount of sensor data and processing power is necessary.

Although **symmetric cryptography** is faster than its counterparts, it requires a secure channel for key exchange, not feasible in IoD's open network environment. Also, key storage and distribution present a challenge in large-scale applications as PQC prompts the use of larger keys. [19]'s **hash**-based lightweight scheme uses AEGIS and SHA-256 to authenticate IoD's time-synchronized nodes. A **fuzzy extractor** is also employed for the user's bio-metric verification, but this scheme does not provide quantum-resistant. A **context-aware** mutual authentication protocol (CoMAD) was designed for swarms in ad-hoc drone networks [6]. This approach uses a master drone for authentication and offers capture resilience and group authentication without relying on infrastructure, secure channels, or secure storage. CoMAD was the first protocol to use context information in re-shaping the swarms. They do not, however, account for user-drone communications.

Public Key Infrastructure (PKI), or **asymmetric cryptography**, does not require a secure channel; however, drones' resource constraints make the direct application of complex schemes like RSA, infeasible. A certificate-less pairing-free aggregate authentication scheme was suggested in [20]. This scheme uses Elliptic-Curve Cryptography (ECC) for authentication between clusters of UAVs and the control center, mitigating the key escrow problem and the certificate management challenge. However, ECC is not quantum-safe. Identity-based and aggregate signature-based authentication schemes were developed to assure confidentiality and data integrity. [21]'s efficient PKI-based framework uses a centralized database, certificate exchanges, and asymmetric encryption to authenticate drones in a fly zone. This solution adopts a lightweight binary certificate, making authentication approximately 3.1 times faster than the "TLS for IoT" protocol. The lightweight AKE protocol for IoD suggested by [22] does not provide forward secrecy and is prone to stolen verifier, privileged insider, user impersonation, and login without verification attacks. [23] proposed a PKI-based key agreement scheme for IoD using dynamic session keys with efficient performance.

It should be noted that the legitimate certificates used in PKI can be exploited by malicious entities. To overcome this drawback, [24] proposed an identity and location validation scheme. In addition to employing the PKI principle, this method authenticates the drones by observing their location and trajectory to detect anomalies in drone mobility patterns. An ECC-based authentication scheme was developed by [25] with comparatively fast modular arithmetic operations. Hash, XOR, RSA, and Fuzzy verifier operations are used by [26] for IoD's data transmission security. However, this approach was proven unsafe against user tracking, drone capture, and impersonation attacks by [8], who enhanced their scheme by incorporating dynamic identities.

The aforementioned approaches do not consider quantum security. The main candidates for **PQC** standardization rely on the hardness of lattice problems such as Learning with Errors (LWE) [9]. However, due to small errors between the computed keys, the direct LWE analogy of the Diffie-Hellman (DH) key exchange is ineffective. To address these inconsistencies, **error reconciliation** is used in a quantum-safe key exchange protocol by [27]. Their method generates the **Ring-LWE** instances by using odd modulus and even errors; allowing two parties to extract the same bit from two similar values (hints). Building on [27]'s work, some adaptable and provably secure schemes have been proposed akin to the existing DH-based protocols, such as [28]'s password-authenticated key exchange (**PAKE**) protocol. In [29], Peikert provided a novel reconciliation mechanism by using even modulus and extracting the key's most significant bit as the common value. The **LWE**-based Frodo [30], generalizes [29]'s mechanism to agree on multiple bits from a single coefficient, at the cost of reduced error tolerance.

An **MLWE**-based **three-party PAKE** scheme was proposed in [31] to address password storage and management in multi-user settings, while also mitigating signal leakage attacks on [29]'s mechanism. A quantum-safe AKE protocol for drones was proposed by [32], using timestamps and RLWE-based cryptography. Despite being efficient, their scheme is susceptible to impersonation, login without verification, and stolen verifier attacks. Furthermore, their failure to incorporate pseudo-identities

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF APPROACHES, FEATURES, AND MITIGATED ATTACKS IN VARIOUS SCHEMES

Scheme	Approach	Security Model	Formal Verification	Dynamic Network	Quantum Safety	P1	P2	P3	P4	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
[22]	Hash	ϕ	BAN,Scyther	ϕ	✗	✓	✗	ϕ	✗	✓	✗	✓	✓	✓	✓	ϕ	✗	ϕ	ϕ
[19]	AEGIS	DY	Scyther	ϕ	✗	✓	✓	✓	✓	ϕ	ϕ	✓	✓	✓	✓	✓	✓	✓	✓
[25]	ECC	DY, CK	Scyther	✓	✗	✓	✓	✓	✓	ϕ	ϕ	✗	✓	✓	✓	✓	✓	✓	ϕ
[35]	ECC	eCK	ϕ	✓	✗	✓	✓	✓	✓	ϕ	✗	✓	✓	✓	✓	✓	✓	ϕ	✓
[36]	ECC	eCK	BAN	ϕ	✗	✓	✓	✓	✓	ϕ	✗	✓	✓	✗	✗	✓	✓	✗	ϕ
[13]	PUF	DY, CK	AVISPA	ϕ	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
[26]	RSA	ϕ	AVISPA	ϕ	✗	✓	✗	✗	✗	ϕ	ϕ	ϕ	✓	✗	✓	✗	✗	ϕ	✓
[8]	MD5	ϕ	BAN	ϕ	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
[32]	RLWE	ϕ	ϕ	ϕ	✓	✓	✓	✓	✗	✓	✓	✓	✓	✗	✗	✓	✗	✓	✓
Ours	**MLWE**	**DY**	**AVISPA**	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note: A1: Eavesdropping; A2: Tampering/spoofing; A3: Replay; A4: MITM; A5: Drone capture; A6: Stolen verifier; A7: Privileged insider; A8: Impersonation; A9: Ephemeral secret leakage; A10: Brute-force; ✓: Supported; ✗: Not supported; ϕ: Not mentioned.

for drones, makes their protocol vulnerable to drone tracking and drone capture attacks.

In summary, prior works either contain vulnerabilities, lack quantum safety, or are not tailored to drone networks. The only known drone-specific PQC-based AKE [32] is vulnerable, lacks formal verification, NIST compliance, and dynamic support. Our protocol addresses these gaps through a comprehensive, standardized, defense-in-depth design. Furthermore, a detailed comparison of security features and methods is provided in Table I.

III. PRELIMINARIES

This section provides essential background knowledge on lattice-based cryptography, key encapsulation, and fuzzy extractors.

A. Lattice-Based Cryptography

Due to LWE's substantial public-key sizes, more efficient variants like Ring-LWE and Module-LWE problems were introduced to optimize lattice-based key exchange mechanisms. Although MLWE introduces a tiny dimension in the polynomial ring structure, it enhances security and design flexibility [29].

Sampling Random Variables on Lattice: A central binomial distribution (CBD) on lattices is used to ensure indistinguishability and improve the efficiency of sampling [31]. This can also be referred to as the error distribution in LWE problems. Let $\mathcal{D}_\eta$ denote the approximate Gaussian distribution on R_q parameterized by η. To sample a polynomial $f \in R_q$ from $\mathcal{D}_\eta$, denoted as $f \leftarrow \mathcal{D}_\eta$, each of f's coefficients is independently sampled. The sampled vector v composed of k polynomials f is denoted as $v \leftarrow \mathcal{D}_\eta^k$.

Module-LWE: Let $\mathbb{Z}_q$ denote the quotient ring $\mathbb{Z}/q\mathbb{Z}$, i.e., the ring of cosets $x + q\mathbb{Z}$ with the induced addition and multiplication operations for an integer $q \geq 1$. Consider the irreducible polynomial $f(x) = x^n + 1$ and the integer polynomial ring $R_q^n = \mathbb{Z}_q[x]/f(x)$. The MLWE problem with parameters (n, q, k, η) denoting the polynomial degree $n = 256$, modulus $q = 3329$, polynomial matrix and vector dimension k, and the standard deviation $\eta \in \{2, 3\}$ of $\mathcal{D}_\eta$ on R_q^n is defined as follows [9]:

Definition 3.1 (MLWE problem): Given (A, y), where the polynomial matrix $A \leftarrow R_q^{k\times k}$, the secret vector $s \leftarrow \mathcal{D}_\eta^k$, and the error vector $e \leftarrow \mathcal{D}_\eta^k$ are chosen uniformly at random, and $y = As + e$ for $y \in R_q^k$; the underlying assumption in the MLWE problem states that finding s is computationally difficult.

Definition 3.2 (Decision-MLWE problem): Given (A, y), the underlying assumption in the decision-MLWE problem states that it is difficult to distinguish whether $y = As + e \in R_q^k$ for uniformly random vectors s and e and matrix A, or $y \leftarrow R_q^k$ is chosen uniformly at random.

Error Reconciliation: Reconciliation mechanisms address the challenge of direct key establishment caused by the structure of hard lattice problems. Due to its efficiency, Peikert's reconciliation mechanism [29] is widely used in the design of LWE-based AKE protocols, reducing the communication costs as well as the probability of agreement failure. For this purpose, the following distance functions are defined [28]:

Lemma 3.1: The distance between two elements $a, b \in \mathbb{Z}_q$ is defined as $|a - b|_q = \min\{(a - b) \mod q, \quad (b - a) \mod q\}$.

Lemma 3.2: The distance between two approximate polynomials $a = \Sigma_{i=0}^{n-1} a_i X^i$ and $b = \Sigma_{i=0}^{n-1} b_i X^i$ is defined as $|a - b|_q = \max\{|a_1 - b_1|_q, |a_2 - b_2|_q, \ldots, |a_{n-1} - b_{n-1}|_q\}$.

In this work, we consider such distances caused by small noises to be relatively small compared to q. In Peikert's method, for $\lfloor x \rceil = \lfloor x + \frac{1}{2} \rfloor \in \mathbb{Z}$ and $E = [-\frac{q}{8}, \frac{q}{8}) \cap \mathbb{Z}$ where $q > 2$ is a prime number; the set $\mathbb{Z}_q = \{-\frac{q}{2}, \ldots, 0, \ldots, \frac{q}{2} - 1\}$ is divided into four intervals: $I_0 = \{0, 1, \ldots, \lfloor \frac{q}{4} \rceil - 1\}$, $I_1 = \{-\lfloor \frac{q}{4} \rfloor, \ldots, -1\}$, $I_0' = I_0 + \frac{q}{2}$, and $I_1' = I_1 + \frac{q}{2}$. For any $x \in \mathbb{Z}_q$, Peikert's modular rounding function $Mod_2(.) : \mathbb{Z}_q \to \mathbb{Z}_2$, is defined as

$$Mod_2(x) = \lfloor \frac{2}{q} x \rceil \mod 2. \tag{1}$$

The hint function $hint(.) : \mathbb{Z}_q \to \mathbb{Z}_2$, also known as the signal, characteristic, or cross-rounding function, is defined as

$$hint(x) = \left\lfloor \frac{4}{q} x \right\rfloor \mod 2 = \begin{cases} 0 & \text{if } x \in I_0 \cup I_0', \\ 1 & \text{if } x \in I_1 \cup I_1'. \end{cases} \tag{2}$$

The reconciliation function $Rec(.) : \mathbb{Z}_q \times \mathbb{Z}_2 \to \mathbb{Z}_2$ is defined as

$$Rec(v, h) = \begin{cases} 0 & \text{if } v \in I_h + E \mod q, \\ 1 & \text{otherwise.} \end{cases} \tag{3}$$

These functions adhere to some fundamental properties outlined in the following lemmas, indicating the correctness and security of the mechanism [29]:

Lemma 3.3: For even modulus q, if $x \in \mathbb{Z}_q$ and $e \in E$, then $Rec(x + e, hint(x)) = Mod_2(x)$.

Lemma 3.4: For even modulus q, if $x \in \mathbb{Z}_q$ is uniformly random, then $Mod_2(x)$ is uniformly random given $hint(x)$. Consequently, the output distribution of $Rec(x + e, hint(x))$ is statistically close to uniform as well.

Peikert used his reconciliation mechanism to construct an IND-CPA secure RLWE-based KEM [29]. The modulus q of the polynomial ring is usually an odd number, causing bias in the reconciled bit's distribution. To address this problem,

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Error Reconciliation Mechanism.

Con(x): $\triangleright\ x \in \mathbb{Z}_q^n$ (odd q)
$\bar{x} = dbl(x)_{\bar{e}} = 2x - \bar{e} \mod 2q$ $\triangleright\ \bar{x} \in \mathbb{Z}_{2q}$
$h = hint(\bar{x}) = \lfloor \frac{4}{q}\bar{x} \rfloor \mod 2$
$k = Mod_2(\bar{x}) = \lfloor \frac{2}{q}\bar{x} \rceil \mod 2$
return (k, h) $\triangleright$ shared key and hint
Rec$(2v, h)$: $\triangleright\ 2v \in \mathbb{Z}_{2q}^n$, $h \in \mathbb{Z}_2^n$, $e \in E$, $\bar{e} \in \mathbb{Z}$
$\bar{x} = 2v - (2e + \bar{e}) \mod 2q$
$k' = \lfloor \frac{2}{2q}\bar{x} \rceil \mod 2$ $\triangleright$ checks $2v \in I_h + E \mod 2q$
return k' $\triangleright$ shared key $(k = k')$

Peikert used the randomized doubling function $dbl(.) : \mathbb{Z}_q \to \mathbb{Z}_{2q}$, defined as $dbl(x) = 2x - \bar{e} \mod 2q$, where for the small random term $\bar{e} \in \mathbb{Z}$, it holds $\Pr[\bar{e} = 0] = \frac{1}{2}$, and $\Pr[\bar{e} = 1] = \Pr[\bar{e} = -1] = \frac{1}{4}$.

Lemma 3.5: For an odd modulus q, if $x \in \mathbb{Z}_q$ is uniformly random and $\bar{x} = dbl(x) \in \mathbb{Z}_{2q}$, then $Mod_2(\bar{x})_{2q}$ is uniformly random on $\mathbb{Z}_{2q}$ given $hint(\bar{x})$.

Consequently, for $x, v \in \mathbb{Z}_q$ and some small $e \in E$, if $v = x + e \mod q$, then $\bar{x}, 2v \in \mathbb{Z}_{2q}$ are also sufficiently close and $Mod_2(\bar{x})_{2q} = Rec(2v, hint(\bar{x})_{2q})$. Extending these functions to the polynomial ring elements on MLWE is the same as performing reconciliation for each polynomial term [31]. The proposed conciliation function $Con()$ and reconciliation function $Rec()$ based on Peikert's mechanism [29] is outlined in Algorithm 1.

PWE: To facilitate the security proof of our proposed scheme, we use the Pairing with Error (PWE) and Decision-PWE problems, which can be reduced to the MLWE problem [28], [31]. Consider $\mathcal{A}$ is a probabilistic polynomial-time (PPT) adversary, $\mathbb{B} = \{0, 1\}$, and A^T denotes the matrix transpose of A.

Definition 3.3 (PWE problem): Given (A, x, y, h), where $A \leftarrow R_q^{k \times k}$, $x \leftarrow R_q^k$, $y = A^T s + e \in R_q^k$, and $(\mathcal{K}, h) = Con(x^T s + e_\nu)$ for $s, e \leftarrow \mathcal{D}_\eta^k$ and $e_\nu \leftarrow \mathcal{D}_\eta$; the goal of $\mathcal{A}$ is to obtain the value of $\mathcal{K}$. The advantage of $\mathcal{A}$ based on the module-PWE assumption is defined as $Adv_{R_q}^{PWE}(\mathcal{A}) = \Pr[\mathcal{K} \in \mathcal{A}(A, x, y, h)]$. Suppose $Adv_{R_q}^{PWE}(t, N) = \max_{\mathcal{A}}\{Adv_{R_q}^{PWE}(\mathcal{A})\}$, where t is the maximum time complexity within which adversaries try to exploit the maximum attack. Adversaries output a list containing at most N elements belonging to $\mathbb{B}^n$. The PWE assumption asserts that $Adv_{R_q}^{PWE}(t, N)$ is negligible, given the security parameter bounds for the polynomial t and N.

Definition 3.4 (Decision-PWE problem): For some $\nu \in R_q$, given $(A, x, y, h, \mathcal{K}) \in R_q^{k \times k} \times R_q^k \times R_q^k \times \mathbb{B}^n \times \mathbb{B}^n$, where $(\mathcal{K}, h) = Con(\nu)$; the Decision-PWE assumption asserts that distinguishing $y = A^T s + e$ and $\nu = x^T s + e_\nu$ for some vectors $s, e \leftarrow \mathcal{D}_\eta^k$, and $e_\nu \leftarrow \mathcal{D}_\eta$, from $(\nu, y) \in R_q \times R_q^k$ chosen uniformly at random is computationally difficult.

Similar to the DMLWE and MLWE problems, it is obvious that if solving DPWE is computationally hard, so is solving PWE.

B. Key Encapsulation Mechanism

The proposed MLWE-based KEM incorporates the aforementioned concepts to enable the secure transmission of a symmetric key over an insecure channel using public keys. This KEM, as outlined in Fig. 1, consists of three main functions:

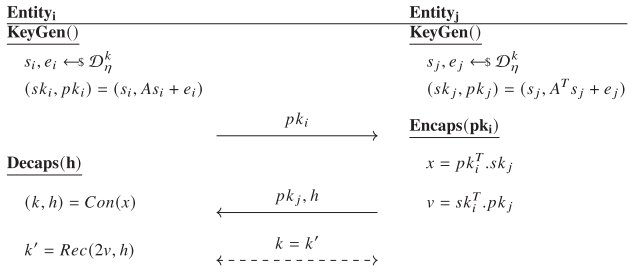


Fig. 1. Key encapsulation mechanism (KEM).

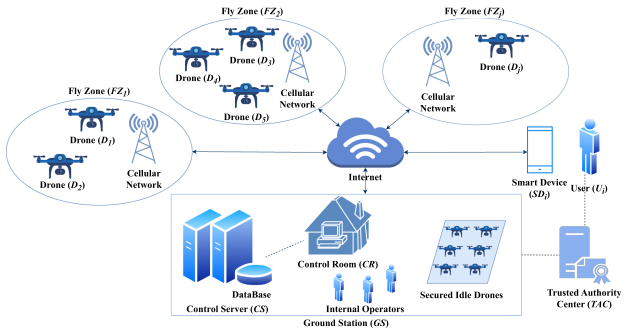


Fig. 2. The drone network model.

key generation to produce a public-private or encryption-decryption key pair (pk, sk), **encapsulation** to generate a symmetric key (k) and its corresponding ciphertext (h), and **decapsulation** to retrieve the symmetric key. NIST's emphasize on KEM over public key encryption is due to its internal key generation functionality; ensuring efficiency and mitigating arbitrary key selection during attacks [9], [31]. Each encapsulation uses fresh randomness to ensure session uniqueness, aligning with NIST's recommendations and mitigating key reuse attacks that can expose internal secrets in lattice-based designs [33].

C. Fuzzy Extractor

In this work, we employ a fuzzy extractor for the bio-metric verification of users and secure storage. The fuzzy extractor FE, has two functions: $(fs_u, fp_u) = FE.Gen(B)$ and $fs_u = FE.Rep(B', fp_u)$, where fs and fp are B's generated secret and public values. fs_u can be reproduced if the Hamming distance of B' from B is less than the error tolerance [19].

IV. System Model

This section outlines the system and security models, detailing entity interactions, assumptions, and threats.

A. Network Model

The drone network studied in this work, as depicted in Fig. 2, essentially consists of a user (U_i) with a smart device (SD_i), a drone (D_j), and a control server (CS). The airspace is divided into multiple flight zones (FZ) in which drones can be deployed to collect data or perform tasks. The server, located at the ground station (GS), stores the relevant information in its secure database, including the data of users and drones. An internal operator is usually present in the control room (CR) to monitor the network. A trusted authority center (TAC) initially registers users in the system and issues their certificates. Technologies like cellular networks can be used to provide wireless connectivity inside a flight zone. Generally, external users need to collect

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

some real-time information directly from the drones rather than using buffered information at the server. To obtain some data or carry out a task, users and drones must authenticate each other with the help of the control server. Additionally, communicating users and drones are required to establish a secure session key. Each entity maintains credentials for authentication and communication, as detailed in Section V: the server stores everyone's $\{ID, PID, T, SID\}$; users store $\{ss_i, f_p, V_u, T_u, PID_u\}$; and drones store $\{ID_d, PID_d, T_d, SID_d\}$. Building upon the approach of CoMAD [6], we consider two operational states for drones during missions: (I) normal and (II) authentication. Various UAV functionalities (e.g., attribute measurement or imagery) are considered normal operations. Whenever an authentication request is received, normal operations are suspended until the authentication process concludes (by acceptance or rejection). For integrity verification of drones during missions, we employ context information, such as flight altitude, visited checkpoints, location, and sensor data (e.g., temperature, pressure, or humidity). It is suggested to use multiple unpredictable context factors for additional security. Context servers or other drones within a swarm can provide and monitor the context data.

B. Threat Model

We adopt the widely accepted Dolev-Yao (DY) threat model [11], in which entities communicate through public (insecure) channels and endpoints are trusted. Adversaries can eavesdrop, forge, or modify the transmitted data. Given drones' deployment in hostile environments, they are vulnerable to physical compromises like capture or memory extraction (e.g., via power analysis). Smart devices may also face theft or side-channel attacks. Server compromise (e.g., privileged insider) is possible but beyond this work's scope.

C. Security Model

This section outlines our security assumptions based on the Bellare-Pointcheval-Rogaway (BPR) security model [10]. This model's support for real-world threats (e.g., forward secrecy and server compromise) makes it suitable for adversarial mission-critical settings like drone networks. The proposed scheme's security is proven via the sequence-of-games approach [34] under the Random Oracle Model (ROM), consistent with BPR and aligning with prior PQC works [28], [31]. Consider a protocol instance $\prod_P^i$ denotes the $i-$th instance of the entity $P \in \{U, D, CS\}$.

Security Game: Participants of an algorithmic game $\mathcal{G}$ include a challenger CH and a PPT adversary $\mathcal{A}$. Essentially, CH runs $\prod_P^i$ on behalf of honest entities, thereby simulating the protocol for $\mathcal{A}$.

Adversary Model: A PPT adversary $\mathcal{A}$ is assumed to have complete control over the communication channel. $\mathcal{A}$ can create, forward, or modify messages, generate multiple instances to participate in the protocol's concurrent execution, and query any given participant instance. Formally, $\mathcal{A}$ is a probabilistic algorithm with a distinct query tape, launching attacks in the real world via random queries. The accepted queries and other security notions aligned with the BPR model are as follows:

- $Execute(\prod_P^i, \prod_Q^j)$: This query characterizes the passive attack capability of $\mathcal{A}$ in obtaining any information transmitted during the protocol's honest execution between $\prod_P^i$ and $\prod_Q^j$.
- $Send(\prod_P^i, m)$: This query characterizes the dynamic attack capability of a PPT adversary. $\mathcal{A}$ sends the generated message m to $\prod_P^i$, who responds back as per the protocol's specifications.
- $Reveal(\prod_P^i)$: This query represents the ability to extract session keys. If $\prod_P^i$ is in the acceptance state with its partner (has received the last anticipated message) and holds the session key SK, this query outputs SK to $\mathcal{A}$.
- $Corrupt(\prod_P^i :)$ By providing the PPT adversary unrestricted access to exploit the entities, this query challenges the forward security of the scheme. If $P = U$, all the information stored on the user's smart device is exposed to $\mathcal{A}$. If $P = D$, the drone's secret values are accessible for $\mathcal{A}$. If $P = CS$, the server's secrets and the data stored in its database are available to $\mathcal{A}$.
- $Test(\prod_P^i)$: This query assesses the PPT adversary's ability to distinguish a legitimate session key from random values. When $\prod_P^i$ receives this query, P chooses a random bit b: if $b = 1$, a random binary string is returned; if $b = 0$, the $Reveal$ query is suppressed and $\prod_P^i$ responds with the real negotiated session key SK. Subsequently, $\mathcal{A}$ guesses the bit b' and wins if $b' = b$. The validity of this query's result depends on ensuring the freshness of the session-instance.

Definition 4.1 (Partnership): Consider a user instance $\prod_U^i$ with $< SID_u^i, PID_u^i, SK_u^i >$ and a drone instance $\prod_D^i$ with $< SID_d^i, PID_d^i, SK_d^i >$, where SID^i, PID^i, and SK^i denote the session identifier, partner identifier, and session key, respectively. $\prod_U^i$ and $\prod_D^i$ are partnered, if:

1) They are both in the state of acceptance.
2) They have the same session identifiers ($SID_u^i = SID_d^i$).
3) The identifying partner for $\prod_U^i$ is $\prod_D^i$ and vice versa ($PID_u^i = D_i, PID_d^i = U_i$, and $SK_u^i = SK_d^i$).
4) No other instances except for $\prod_U^i$ and $\prod_D^i$ are in the state of acceptance with the same partner and session.

Definition 4.2 (Freshness): An instance $\prod_P^i$ is considered fresh if it is an honest session and:

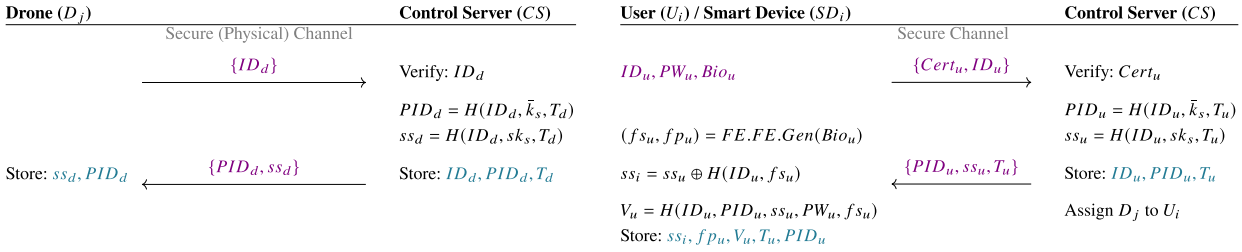
1) $\prod_P^i$ is in the state of acceptance.
2) No $Reveal(\prod_P^i)$ query has been performed on the user instance $\prod_U^i$ or the partner drone instance $\prod_D^i$.
3) No $Corrupt(\prod_P^i)$ query has been performed on $\prod_U^i$, the partner $\prod_D^i$, or the server instance $\prod_S^i$.

Definition 4.3 (Success): Let $Suc(\mathcal{A})$ be the event of $\mathcal{A}$ making a $Test(\prod_P^i)$ query to some fresh session i that has terminated successfully. $\mathcal{A}$ outputs a guessed bit b' corresponding to the $Test$ query's b. If $b = b'$, the adversary is successful; so the probability of $\mathcal{A}$'s success is defined as $\Pr[Suc_{\mathcal{A}}^P(\Pi^i)] = \Pr[b = b']$. Subsequently, the advantage of $\mathcal{A}$ in breaking the security of SK in the polynomial time t is defined as:

$$Adv_{\mathcal{A}}(t) = \left| \Pr[Suc_{\mathcal{A}}^P(\Pi^i)] - \frac{1}{2} \right|. \tag{4}$$

Definition 4.4 (Semantic security): The proposed protocol is secure, provided that instances of an honest user and drone are partnered and generate the same session key (otherwise the protocol has failed). Under the BPR model, the goal of $\mathcal{A}$ is to distinguish the session key from a random value. The advantage

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.



(a) Drone registration process. (b) User registration process.

Fig. 3. The proposed registration processes for drones (a) and users (b) with the control server through secure channels.

of adversary $\mathcal{A}$ in attacking $\prod_P^i$ is defined as:

$$Adv_{\mathcal{A}}^{AKE}(\Pi^i) = |2\Pr[Suc_{\mathcal{A}}^{P}(\Pi^i)] - 1|. \quad (5)$$

The protocol is semantically secure if $Adv_{\mathcal{A}}^{AKE}(\Pi)$ is negligible for all probabilistic polynomial-time adversaries $\mathcal{A}$.

V. Proposed Scheme

This section describes our proposed authentication and communication security scheme, consisting of five processes: initialization, drone/user registration, authentication and session establishment, user operations for updating credentials, and drone operations for dynamic network management. Each subsection details the sequential operations of involved entities, using the notations defined in Section III. Additionally, a proof of correctness is provided to verify reliability and failure rate of the scheme.

A. Initialization

In this phase, the server sets the Fig. 1's KEM instance parameter n, modulus q, matrix A and CBD $\mathcal{D}_\eta^k$. The server generates a private and public key pair $(sk_s, pk_s) = (s_s, As_s + e_s)$, using KEM's key generation function, as well as a semi-static secret $\bar{k}_s$. The server then publishes its public key pk_s, parameters $(n, q, A, \mathcal{D}_\eta^k)$, and hash function $H : \{0,1\}^* \to \{0,1\}^l$ in a public domain.

B. Registration

Registration occurs over secure channels: physical for drones and pre-established key exchanges (e.g., KEMTLS) for users. During this phase, shared secrets ss_u and ss_d are established for server-user and server-drone communications, used for encryption, authentication, and verifying future data transmissions. To preserve anonymity, each user and drone has an ephemeral pseudo-identity PID which masks the entity's identifier ID with the server's semi-static secret $\bar{k}_s$. The temporary credentials of an entity i at time T_i are calculated as $PID_i = H(ID_i, \bar{k}_s, T_i)$ and $ss_i = H(ID_i, sk_s, T_i)$. In addition to providing forward security, the limited validity period of credentials also reduces the window of ephemeral secret leakage attacks. The control server stores $\{ID_i, PID_i, T_i\}$ in its secure database and returns $\{PID_i, ss_i, T_i\}$ to the entity. As outlined in Fig. 3, the registration process for drones (3a) and users (3b) with the server is as follows:

Drone Registration: Drones are assumed to be kept at a secure location (e.g., ground station) prior to deployment and are only under the control and access of legitimate operators. These operators can install mission-specific information, set the network topology, and configure the drones accordingly.

Algorithm 2: User Authentication Process on the Smart Device.

procedure UserAuth (ID_u, PW_u, Bio_u)
  $fs_u = FE.Rep(Bio_u, fp_u)$
  $ss_u = ss_i \oplus H(ID_u, fs_u)$
  $V_i = H(ID_u, PID_u, ss_u, PW_u, fs_u)$
  ensure $V_u = V_i$
end procedure

A drone D_j is registered based on the information provided by its manufacturer and an identifier ID_d. The drone stores $\{ss_d, PID_d\}$ in its memory.

User Registration: A legitimate user U_i with a smart device SD_i, is assumed to have an identifier ID_u and a certificate $Cert_u$ issued by TAC. The server verifies $Cert_u$ with TAC upon receiving a registration request. To mitigate device-based attacks (e.g., power analysis, login without verification, or stolen verifier), ss_u is not directly stored in SD_i's memory. Instead, this value is encrypted with the user's chosen password PW_u and bio-metric information Bio_u by employing a Fuzzy Extractor. ss_u is masked as $ss_i = ss_u \oplus H(ID_u, fs_u)$, where fs_u and fp_u are the secret bio-metric key and the public reproduction parameter in $(fs_u, fp_u) = FE.Gen(Bio_u)$, respectively. A verification parameter $V_u = H(ID_u, PID_u, ss_u, PW_u, fs_u)$ is defined for future user authentications and $\{ss_i, fp_u, V_u, T_u, PID_u\}$ are stored by SD_i.

C. Authentication and Session Establishment

This process, as depicted in Fig. 4, is initiated by the user (U_i) through the smart device (SD_i). The user logs in with the control server (CS), who then sends a deployment command to the corresponding drone (D_j). Finally, a session key (SK) is established between the user and the drone. Upon receiving a message with the timestamp T at the current time T^*, each entity first checks the freshness of the message by verifying $|T^* - T| < \Delta T$. The maximum transmission delay ΔT is the threshold value restricting replay attacks. Given the ephemeral nature of credentials, session keys, and authentication values, which are also bound to timestamps, this process is repeated periodically to maintain integrity throughout prolonged sessions. Our use of ephemeral values and randomized encapsulation also prevents key reuse attacks.

User Authentication: As per Algorithm 2, upon inputting $\{ID_u, PW_u, Bio_u\}$, the user is authenticated by the smart device using the stored parameters $\{ss_i, fp_u, V_u\}$. The user then proceeds to exchange a secret key with the server to log into the network.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

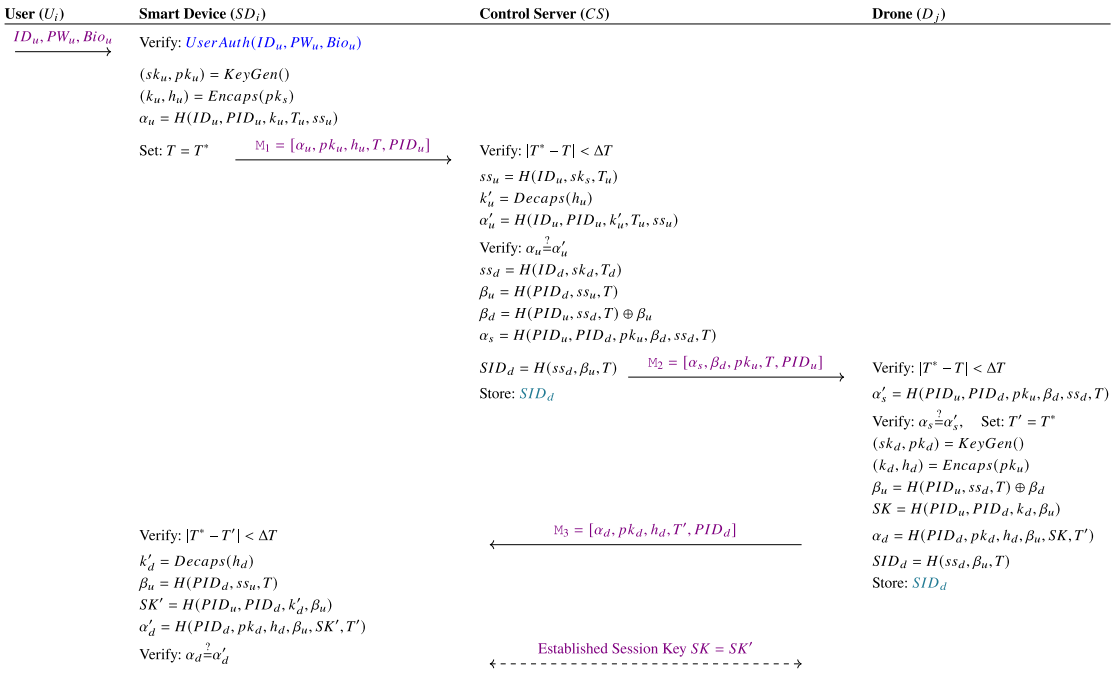


Fig. 4. The proposed user authentication, login, drone deployment, and session establishment process.

Login: The MLWE-based KEM is employed to derive the user's hint value h_u and the conciliated shared secret k_u. For the user's generated private and public keys $(sk_u, pk_u) = (s_u, A^T s_u + e_u)$ and $x_u = pk_s^T.sk_u$, the server's public key pk_s is encapsulated as $(k_u, h_u) = Encaps(pk_s)$. The user's authentication parameter for the server is $\alpha_u = H(ID_u, PID_u, k_u, T_u, ss_u)$. The smart device sends $\mathtt{M}_1 = [\alpha_u, pk_u, h_u, T, PID_u]$ with the fresh timestamp $T = T^*$ to the server. After verifying $|T^* - T| < \Delta T$, the server retrieves user's $\{ID_u, PID_u, T_u\}$ from its secure database and computes $ss_u = H(ID_u, sk_s, T_u)$. Using the server's private key sk_s and user's public key pk_u as $v_u = sk_s^T.pk_u$, the reconciliated shared secret $k'_u = Decaps(h_u)$ is derived by decapsulating h_u. If successful $(k_u = k'_u)$, for $\alpha'_u = H(ID_u, PID_u, k'_u, T_u, ss_u)$, we have $\alpha_u = \alpha'_u$; thereby $\mathtt{M}_1$ is verified, and the user is authenticated.

Drone Deployment: Similarly, the drone's $\{ID_d, PID_d, T_d\}$ are retrieved and $ss_d = H(ID_d, sk_d, T_d)$ is computed. The server then uses $\beta_u = (HPID_d, ss_u, T)$ and $\beta_d = (HPID_u, ss_d, T \oplus \beta_u)$ to correlate session requests with the respective users and drones. This helps avoid replay and impersonation attacks. The server's authentication parameter is $\alpha_s = H(PID_u, PID_d, pk_u, \beta_d, ss_d, T)$. The server sends $\mathtt{M}_2 = [\alpha_s, \beta_d, pk_u, T, PID_u]$ to D_j and instructs it to initiate a new session, identified by $SID_d = H(ss_d, \beta_u, T)$, with U_i. After verifying $|T^* - T| < \Delta T$, the drone calculates $\alpha'_s = H(PID_u, PID_d, pk_u, \beta_d, ss_d, T)$ using the values stored in its memory to confirm the message's integrity $(\alpha_s = \alpha'_s)$.

Session Establishment: This process is secured by using KEM and server-based attacks on the session (e.g., key escrow attack) are mitigated by excluding the server. For the drone's generated private and public keys $(sk_d, pk_d) = (s_d, As_d + e_d)$ and $x_d = pk_u^T.sk_d$, the user's public key pk_u is encapsulated, resulting in the shared secret and hint values $(k_d, h_d) = Encaps(pk_u)$. Using $\beta_u = H(PID_u, ss_d, T \oplus \beta_d)$, the drone computes the session key $SK = H(PID_u, PID_d, k_d, \beta_u)$. The drone's authentication parameter is $\alpha_d = H(PID_d, pk_d, h_d, \beta_u, SK, T')$ with a fresh timestamp T'. Subsequently, the drone sends $\mathtt{M}_3 = [\alpha_d, pk_d, h_d, T', PID_d]$ directly to the user and stores the session identifier $SID_d = H(ss_d, \beta_u, T)$. After checking the message transmission delay $(|T^* - T'| < \Delta T)$, the user (smart device) authenticates the drone and verifies $\mathtt{M}_3$. The reconciliated shared secret $k'_d = Decaps(h_d)$ is derived by decapsulating h_d, using the drone's public key pk_d and user's private key sk_u as $v_d = sk_u^T.pk_d$. The session key $SK' = H(PID_u, PID_d, k'_d, \beta_u)$ is derived using k'_d and $\beta_u = H(PID_d, ss_u, T)$. For $\alpha'_d = H(PID_d, pk_d, h_d, \beta_u, SK', T')$, if $\alpha_d = \alpha'_d$, the symmetric session key $SK = SK'$ is accepted and used to encrypt the subsequent drone-user communications.

D. User Operations

Drone networks face dynamic events such as node addition, removal, or re-engagement, whether planned (e.g., expiration or separation) or unplanned (e.g., accidents, faults, attacks, etc). To address the extensive nature of events, this work defines the following secured user operations for users, as even minor inconsistencies can severely impact other operations or the network.

Password/Bio-metric Update: A comprehensive scheme should enable local password and bio-metric updates without server interactions. [32]'s design for this operation requires the smart device to access the server's secret key, making their protocol prone to login without verification and stolen verifier attacks. Our scheme adopts a secure update process similar to User Registration. Upon the user inputting $\{ID_u, PW_u, Bio_u\}$ and successful authentication per Algorithm 2, the new Bio'_u and PW'_u are employed to mask ss_u. Therefore, for $(fs'_u, fp'_u) = FE.FE.Gen(Bio'_u)$, the values of fp'_u, $V'_u = H(ID_u, PID_u, ss_u, PW'_u, fs'_u)$, and $ss'_i = ss_u \oplus H(ID_u, fs'_u)$ are updated in the device's memory.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

Re-Issue: Legitimate users should be able to securely replace their smart devices or renew expired credentials, as insecure handling may lead to impersonation attacks [19]. The user's certificate and identifier are encrypted as $\langle Cert_u, ID_u\rangle_K$ using either the expired $K = ss_u$ or a new conciliated $K = k_u$, depending on security requirements. Upon verifying $\mathtt{M}_{reissue} = [\langle Cert_u, ID_u\rangle_K, PID_u, T]$, the server expires the user's credentials and issues new ones. $[\langle PID'_u, ss'_u, T^*_u\rangle_K, T_s]$ is sent back to the smart device which then masks and stores the new credentials, similar to User Registration.

E. Drone Operations

To enhance security and prevent unauthorized access, this scheme supports secure dynamic operations and leverages context data to address drone networks' intermittent connectivity, node mobility, dynamic topologies, and unsupervised environments.

Synchronization: Periodic integrity verification of drones mitigates drone capture and side-channel attacks by narrowing the attack window, while also detecting sensor manipulations and localization errors. The drone's ID_d, session's SID_d, and context factors $C = (C_1 \oplus C_2 \oplus \cdots \oplus C_n)$ are used to calculate $CInfo_d = H(C, ss_d, ID_d, SID_d, T)$ at time T. Drone impersonation and de-authentication attempts are mitigated by employing ss_d. The drone sends a secure synchronization message $\mathtt{M}_{sync} = [CInfo_d, PID_d, T]$ periodically, allowing the server to verify its integrity. Failed verification leads to the Dynamic Revocation of drone's access. Furthermore, we can consider different synchronization periods for a node based on its operational state; drones idle at the ground stations require fewer synchronizations, whereas active drones in the field necessitate frequent updates.

Dynamic Addition: Drones may rejoin the network after incidents or during resource transfers. While pre-deployment drones are trusted, newly joining ones pose infiltration risks and cannot be assumed secure. During this process, the drone sends a join request to the control server. If the drone's integrity is verified per Synchronization, new temporary credentials $\{PID'_d, ss'_d, T'_d\}$ are issued. SID_d can be omitted from $CInfo_d$ if not available for new drones. Given the short-term validity of these credentials, storing outdated context data is not necessary for the server since the drone's access is automatically revoked after a prolonged absence.

Dynamic Revocation: An unplanned drone departure (e.g., due to failure or attack) halts communications, necessitating proactive policies for data wipe upon impact or unscheduled landings. Alternatively, drones may be forced out of the network due to synchronization/authentication failure or leave voluntarily (e.g., because of low battery or new tasks) [6]. To prevent impersonation and unauthorized access, the server terminates sessions, expires credentials, and instructs return to the ground station for inspection. To reduce overhead and mitigate DoS, drones notify users of session termination. If a node is malicious however, the server blacklists its credentials and broadcasts this to the network. Note that the revocation request $\mathtt{M}_{revoke} = [H(ID_d, ss_d, T, PID_d, T])$ must be secured and verifiable by the server to prevent malicious interference with legitimate operations. For temporary departures (e.g., battery replacement), drones send a leave threshold T_l to the server. If T_l expires, access is revoked; otherwise, the node is re-authenticated per Dynamic Addition. This also enables energy saving during low-traffic periods via node hibernation.

F. Proof of Correctness

In this section, we prove the protocol's correctness, evidenced by the significant probability of honest participants' negotiated values being equal ($SK = SK'$). For this purpose, $k_d \leftarrow Con(x_d)$ and $k'_d \leftarrow Rec(2v_d, h_d)$ must be equal. Subsequently, correctness depends on the approximate distance between $\bar{x}_d$ and $2v_d$. By definition: $|v_d - x_d|_q = |sk_u^T.pk_d - pk_u^T.sk_d|_q = |s_u^T e_d - e_u^T s_d|_q$. Based on Lemma 3.5, for $\bar{x}_d = 2x_d - \bar{e_d}$ and $e \in [-\frac{q}{8}, \frac{q}{8}) \cap \mathbb{Z}$, where $e = s_u^T e_d - e_u^T s_d$, the distance is equal to $|2v_d - \bar{x}_d|_q = |2v_d - (2x_d + \bar{e_d})|_q = |2e + \bar{e_d}|_q$.

In Peikert's mechanism [29], for the odd modulus q, the error tolerance range is $\lfloor \frac{q}{4} \rfloor$; i.e., if $|2e + \bar{e_d}|_q \leq \lfloor \frac{q}{4} \rfloor$, then $k_d = Mod_2(\bar{x}_d) = Rec(2v_d, h_d) = k'_d$. By definition, the small polynomial vector $\frac{1}{2}\bar{e_d}$ is negligible in $|e + \frac{1}{2}\bar{e_d}|_q \leq \lfloor \frac{q}{8} \rfloor$ and:

$$|e| = |s_u^T e_d - e_u^T s_d|_q \leq \left\lfloor \frac{q}{8} \right\rfloor. \tag{6}$$

The authentication of the user by the server follows a similar pattern; if $|2v_u - \bar{x_u}|_q = |2e' + \bar{e_u}|_q \leq \lfloor \frac{q}{4} \rfloor$, we have $k_u = Mod_2(\bar{x_u}) = Rec(2v_u, h_u) = k'_u$. Therefore:

$$|e'| = |s_s^T e_u - e_s^T s_u|_q \leq \left\lfloor \frac{q}{8} \right\rfloor. \tag{7}$$

Consequently, honest participants can negotiate the same value in the proposed AKE scheme with the failure rate of:

$$\delta = \Pr\left[|s_u^T e_d - e_u^T s_d|_q \geq \left\lfloor \frac{q}{8} \right\rfloor\right] + \Pr\left[|s_s^T e_u - e_s^T s_u|_q \geq \left\lfloor \frac{q}{8} \right\rfloor\right]. \tag{8}$$

VI. Security Analysis

This section presents the security analysis of our scheme, including a security proof and automated formal model verification.

A. Proof of Security

Similar to prior works, we adopt the sequence-of-games technique [34] under the BPR model, ensuring negligible advantage under MLWE hardness and ROM. This proves the security achievements outlined in Section VII-A. Games model real-world adversarial objectives. Let $\mathcal{A}$ be a PPT adversary interacting with instances $\prod_U^i$, $\prod_D^i$, and $\prod_{CS}^i$ in session i. The goal is to demonstrate that $\mathcal{A}$'s advantage in learning the session key SK of a fresh instance is negligible. Consequently, the protocol is secure, if the PW and ID dictionaries, the hash function's range, and the CBD ($\mathcal{D}_\eta$) are sufficiently large, and $\mathcal{A}$ cannot solve the MLWE problem with a significant advantage. Each participant is instantiated as a set of deterministic probabilistic polynomial oracles: an unknown query x returns a randomly chosen y, with (x, y) stored for consistency.

Theorem 6.1: Let $\prod$ be an instance of the proposed AKE protocol and q_j the number of queries made to the oracle $j \in \{test, reveal, corrupt, snd, exe, ro, H\}$ representing the $Test$, $Reveal$, $Corrupt$, $Send$, $Execute$, $Random$, and $Hash$ oracles, respectively. The advantage of a PPT adversary $\mathcal{A}$ in modifying the session key SK within the polynomial time t is denoted as $Adv_{\mathcal{A}}(t)$. The term $Adv_{\mathcal{A}}^{MLWE}(t')$ indicates the adversary's advantage in solving an instance of the MLWE problem

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

within the polynomial time t'. Suppose the emulator controls all the oracles accessible to $\mathcal{A}$, and the simulator runs the protocol. For $t' = \mathcal{O}(t + (q_{exe} + q_{snd} + q_{ro} + q_H))$, the advantage of $\mathcal{A}$ is limited to:

$$Adv_{\mathcal{A}}^{AKE}(t) \leq \frac{q_H^2}{2^l} + \frac{q_K^2}{q^{nk}} + \frac{(q_{ro} + q_{snd} + q_{exe})^2}{q^{nk}} + 2(q_{exe} + q_{snd}).Adv_{\mathcal{A}}^{PWE}(t', q_{ro}) + 2Adv_{\mathcal{A}}^{MLWE}(t', q_{ro})$$

Proof: The adversary $\mathcal{A}$ and the challenger CH play a series of games denoted as $\mathcal{G}_i$ for $i = \{0, \ldots, 7\}$, in which $\mathcal{A}$ will make polynomial-many queries to CH. Suc_i denotes $\mathcal{A}$'s success as the result of the $Test$ queries in $\mathcal{G}_i$. The adversary has access to the publicly transferred messages $\{\mathtt{M}_1, \mathtt{M}_2, \mathtt{M}_3\}$. The sequence of games and their success probability is as follows:

$\mathcal{G}_0$ *(ROM Attack):* This game simulates a real attack against the protocol under ROM; therefore

$$Adv_{\mathcal{A}}(\mathcal{G}_0) = |2\Pr[Suc_0] - 1|. \tag{9}$$

$\mathcal{G}_1$ *(Eavesdropping Attack):* This game simulates $\mathcal{A}$ eavesdropping on public channels. $\mathcal{A}$ obtains the information between honest participants through the $Execute$ oracle in order to decide if the $Test$ output is a valid session key or a random value.

Corollary 6.1: For a PPT $\mathcal{A}$, $\mathcal{G}_0$ and $\mathcal{G}_1$ are indistinguishable:

$$Adv_{\mathcal{A}}(\mathcal{G}_1) = Adv_{\mathcal{A}}(\mathcal{G}_0) \tag{10}$$

Proof 6.1: By monitoring the public channel, $\mathcal{A}$ can obtain $\{\alpha_u, \alpha_s, \alpha_d, pk_u, pk_d, h_u, h_d, \beta_d, T, T', PID_u, PID_d\}$; but $\mathcal{A}$ does not know ss_d and cannot generate k_d or β_u to drive the session key. Therefore, $\mathcal{G}_0$ and $\mathcal{G}_1$ are indistinguishable and this attack does not increase the probability of $\mathcal{A}$'s success ($\Pr[Suc_0] = \Pr[Suc_1]$).

$\mathcal{G}_2$ *(Hash Collision):* In this game, CH simulates a random oracle H_k by maintaining a hash list $\mathcal{H}_k$. The adversary queries the Hash oracle adaptively to identify hash collisions.

Corollary 6.2: For a PPT adversary $\mathcal{A}$:

$$|Adv_{\mathcal{A}}(\mathcal{G}_2) - Adv_{\mathcal{A}}(\mathcal{G}_1)| \leq \frac{q_H^2}{2^{l+1}} + \frac{q_K^2}{2 \times q^{nk}} \tag{11}$$

Proof 6.2: Suppose the KEM instance uses a hash function modeled as the oracle K that outputs a polynomial of degree n and dimension k, where each coefficient can have q possible values. As per the birthday paradox, the probability of collision is less than $\frac{q_H^2}{2\times|H|} + \frac{q_K^2}{2\times|K|}$, where q_H and q_K are the number of queries to H and K with the range space of 2^l and q^{nk}, respectively.

$\mathcal{G}_3$ *(Random Collision):* This game simulates collisions in the key derivation function (KDF), allowing $\mathcal{A}$ to use an ideal cipher key that corresponds to multiple inputs. $\mathcal{A}$ could use this value to successfully attack a session, even if an incorrect session key is used.

Corollary 6.3: For a PPT adversary:

$$|Adv_{\mathcal{A}}(\mathcal{G}_3) - Adv_{\mathcal{A}}(\mathcal{G}_2)| \leq \frac{(q_{ro} + q_{snd} + q_{exe})^2}{2 \times q^{nk}} \tag{12}$$

Proof 6.3: Suppose that KDF keeps a list of all prior queries and their responses. If collision occurs, CH aborts and $\mathcal{A}$ fails. The probability of collision in the calculation of $\mathtt{M}_1$, $\mathtt{M}_2$, or $\mathtt{M}_3$ using the $Random$, $Send$, and $Execute$ queries, as per the birthday paradox is at most $\frac{(q_{ro}+q_{snd}+q_{exe})^2}{2\times|\mathcal{D}_\eta^k|}$, where $|\mathcal{D}_\eta^k| = q^{nk}$ is the CBD range space for sampling the secret and error vectors.

$\mathcal{G}_4$ *(Solving MLWE):* In this game, CH simulates an attack on an instance of the MLWE problem to obtain the session key. The value of SK may be calculated without using the hash and random functions; i.e., the user and drone's ephemeral keys are not MLWE-distribution samples but are instead guessed without checking the consistency of the output with the $Execute$ query.

Corollary 6.4: For a PPT adversary:

$$|Adv_{\mathcal{A}}(\mathcal{G}_4) - Adv_{\mathcal{A}}(\mathcal{G}_3)| \leq Adv_{\mathcal{A}}^{MLWE}(t', q_{ro}) + (q_{exe} + q_{snd}).Adv_{\mathcal{A}}^{PWE}(t', q_{ro}) \tag{13}$$

Proof 6.4: If $\mathcal{A}$ can derive SK with non-negligible probability, then a polynomial-time algorithm $\mathcal{Q}$ exists that can solve the PWE problem by simulating $\mathcal{G}_3$ within $t' = \mathcal{O}(t + (q_{exe} + q_{snd} + q_{ro} + q_H))$. The polynomial time t' is the average elapsed time for one component-wise scalar multiplication in R_q. Given (A, x, y, h), upon $\mathcal{A}$ making an $Execute$ query, $\mathcal{Q}$ sets $pk_u = x + A^T.s_u + e_u$ and $pk_d = y + A.s_d + e_d$, where $s_u, e_u, s_d, e_d \leftarrow \mathcal{D}_\eta^k$. For each $SK = H(PID_u, PID_d, k_d, \beta_u)$ query, $k'_d = Rec(2v_d, h_d)$ where $v_d = sk_u^T.pk_d$ corresponds to $x_d = pk_u^T.sk_d$. Then, $\mathcal{Q}$ can compute $\bar{x}_d$ and $\mathcal{K} = Rec(2v_d, h_d) = \lfloor \frac{2}{2q}\bar{x}_d \rceil_2$. Subsequently, $\mathcal{Q}$ adds $\mathcal{K}$ to the list of PWE's possible answers, at which point the PWE problem is solved. When simulating $\mathcal{G}_3$, $\mathcal{Q}$ replaces $pk_u = A^T s_u + e_u$ and $pk_d = As_d + e_d$ with $pk_u = y + A^T s_u + e_u$ and $pk_d = x + As_d + e_d$, respectively. Since $x \leftarrow R_q^k$ is uniformly random and $y = A^T s_y + e_y$, the assigned and actual values of pk_d and pk_u are indistinguishable unless $Adv_{\mathcal{A}}^{DMLWE}$ is non-negligible.

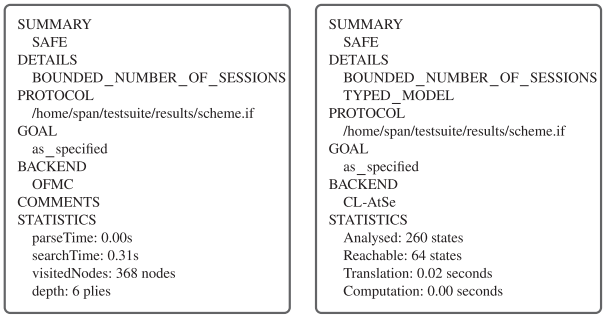
$\mathcal{G}_5$ *(Corrupted Entity):* In this game, CH simulates entity compromises. By corrupting the control server (e.g., privileged insider or key escrow attacks), $\mathcal{A}$ gains access to the registration data sent through secure channels, the server's secrets, and any data stored in the secure database. The user's smart device and the UAV may also be corrupted through theft, malware, or tampering; allowing $\mathcal{A}$ to control these entities and extract secret parameters from their memories through side-channel attacks. Such attacks can lead to bypassing the authentication phase or impersonation.

Corollary 6.5: For a PPT $\mathcal{A}$, $\mathcal{G}_4$ and $\mathcal{G}_5$ are indistinguishable:

$$Adv_{\mathcal{A}}(\mathcal{G}_5) = Adv_{\mathcal{A}}(\mathcal{G}_4) \tag{14}$$

Proof 6.5: Suppose that $\mathcal{A}$ can obtain $\{ID_u, ID_d, T_u, T_d, sk_s, \bar{k}_s\}$ with a $Corrupt(\mathbf{CS})$ query. Consequently, $\mathcal{A}$ can compute $\{ss_u, ss_d, \beta_u, \beta_d\}$ of any user or drone. However, $\mathcal{A}$'s attempt at modifying encrypted messages or compromising sessions is of no avail since calculating SK is equal to solving the MLWE problem as per $\mathcal{G}_4$. Other user-oriented phases of the scheme are also secure, as $\mathcal{A}$ cannot replicate the user's Bio_u or $Cert_u$; therefore, $\mathcal{A}$ can not verify a guessed PW_u or impersonate a legitimate user either. Suppose that $\mathcal{A}$ can obtain $\{ss_i, fp_u, T_u, V_u\}$ with $Corrupt(\mathbf{U})$. However, without knowing fs_u, $\mathcal{A}$ can not derive ss_u. Hence, $\mathcal{A}$ can not verify an online or offline guess of the ID_u and PW_u combination through $Send$ queries or modify α_u. Suppose that $\mathcal{A}$ can obtain $\{ss_d, SID_d\}$ with $Corrupt(\mathbf{D})$. Since shared secrets and session keys are independent, the security of other sessions and entities remain unaffected. Periodic synchronization aids in detecting tampering or capture attacks and reduces the vulnerability window for power analysis. This design also offers resistance against key compromise impersonation (KCI). The corrupted entity attack,

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.



(a) Simulation result of OFMC. (b) Simulation result of CL-AtSe.

Fig. 5. The proposed scheme's analysis using the AVISPA tool.

therefore, does not increase the probability of $\mathcal{A}$'s success, meaning $\mathcal{G}_4$ and $\mathcal{G}_5$ are indistinguishable.

If all of the PPT adversary's attempts at breaking the security of the protocol are unsuccessful and all the games have been executed, then the last endeavor to win would be querying the $Test$ oracle and guessing b. So $Adv_{\mathcal{A}}(\mathcal{G}_5) = \frac{1}{2}$ and as per (5): $Adv_{\mathcal{A}}^{AKE}(t) = |2\Pr[Suc_{\mathcal{A}}^{P}(\Pi)] - 1| = 2|\Pr[Suc_0] - \Pr[Suc_5]|$.

Subsequently, applying the triangular inequality results in

$$Adv_{\mathcal{A}}^{AKE}(t) \leq 2\sum_{i=0}^{4} |\Pr[Suc_i] - \Pr[Suc_{i+1}]|.$$

Finally, according to the (9) to (14):

$$Adv_{\mathcal{A}}^{AKE}(t) \leq 2 \times \left(\frac{q_H^2}{2^{l+1}} + \frac{q_K^2}{2 \times q^{nk}} + \frac{(q_{ro} + q_{snd} + q_{exe})^2}{2 \times q^{nk}} + (q_{exe} + q_{snd}) Adv_{\mathcal{A}}^{PWE}(t', q_{ro}) + Adv_{\mathcal{A}}^{MLWE}(t', q_{ro}) \right);$$

thereby, proving Theorem 6.1. □

B. Security Validation

We verify the security of our protocol's formal model by using the well-established Automated Validation of Internet Security Protocols and Applications (AVISPA) project [12]. We utilize the OFMC and CL-AtSe backends, as they support bit-wise XOR operations. The scheme's registration, and authentication procedures were simulated using SPAN. Fig. 5 shows the OFMC (5a) and CL-AtSe (5b) simulation results. These outcomes indicate that the proposed scheme can be executed securely, without any sign of replay or MITM attacks by passive or active adversaries [12].

VII. Evaluation

This section evaluates our scheme's security, performance, and comparison with existing solutions.

A. Security Achievements

This section presents our scheme's security features, demonstrating robustness against attacks and ensuring confidentiality, integrity, and availability through the following properties:

Mutual Authentication (P1): This property is satisfied between U_i/SD_i, CS, and D_j. Regenerating fs_u with U_i's biometric information authenticates U_i on SD_i. The values of α_u, α_s, and α_d are then used in the authentication of U_i by CS, CS by D_j, and D_j by U_i, respectively. ss_u and ss_d also help in identity verification by CS. The integrity of messages is verified in a similar process by using β_d and β_u. This is formally verified via Games $\{0, 1, 2, 5\}$.

Secure Key Establishment (P2): $\{ss_u, ss_d\}$ are established through secure channels, and $\{k_u, k_d, SK\}$ by employing KEM, hashing, and secret parameters (β_u, β_d). As captured in Games 1–4, the scheme, therefore, guarantees secure key establishment.

Forward Security (P3): Ephemeral values, timestamps, and fresh seeds are used to establish independent shared secrets; thereby maintaining the security of previous sessions even if the keys are compromised. This is covered by Game 5.

Anonymity/Untraceability (P4): The identity of entities are decoupled from their data by employing ephemeral pseudo-identities. Random nonces and timestamps ensure message independence across sessions, preventing activity tracing and ensuring anonymity and untraceability. Games $1, 2, 5$ confirm session indistinguishability and unlinkability, even under partial compromise.

Mitigated Attacks: As detailed in previous sections, our protocol ensures robustness against various attacks on drone networks, including eavesdropping, tampering/spoofing, replay, MITM, tracking, drone capture, stolen verifier, privileged insider, impersonation, ephemeral secret leakage, brute-force, and key reuse attacks. Table I provides a detailed comparison of cryptographic approaches, security models, verification techniques, dynamic operations/mobility support, and security guarantees, including mitigated attacks, across various schemes. It highlights our contributions as well as the vulnerabilities and shortcomings of existing works. Notably, most protocols lack quantum-resistance. Despite [32]'s quantum-safety claims, their scheme is susceptible to drone tracking, drone capture, stolen verifier and impersonation attacks. General PQC schemes are excluded from this comparison, as they do not align with drone-specific security requirements or threat models.

TABLE II
Notations and Execution Costs of Operations (ns) [32], [41], [42]

Symbol	Operation	User/Drone	Server
T_h	Hash function	180.964	14.090
T_{genR}	RLWE key generation	1159.126	149.853
T_{encR}	RLWE encapsulation	36.820	0.996
T_{decR}	RLWE decapsulation	1.305	0.307
T_{genM}	MLWE key generation	1967.555	254.368
T_{encM}	MLWE encapsulation	62.171	1.682
T_{decM}	MLWE decapsulation	2.204	0.518
T_{sym}	Symmetric encryption	7000	2000
T_{ecca}	ECC point addition	78000	12000
T_{eccm}	ECC point multiplication	3437000	417000
T_{fe}	Fuzzy extractor	1434	-
T_{puf}	PUF	16	-

B. Performance Analysis

In this section, we assess the performance of our proposed protocol. To achieve NIST's level 1 security, we use SHA3-512 [37] and ML-KEM-512 [9]. So $H()$'s output is $l = 512$ bits and the encapsulation key (pk), decapsulation key (sk), ciphertext (h), and shared secret key (k) are 800, 1632, 768, and 32 bytes, respectively. Consequently, SK, V_u, ss, α, and β are also 512 bits. We consider ID, PW_u, fp_u, $\bar{k}_s$, and T to be 32 bits each. Enhancing security often undermines the performance; however, we propose the use of a 32-bit hash

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF THE STORAGE (BITS), COMMUNICATION (BITS), AND EXECUTION (NANOSECONDS) COSTS IN VARIOUS SCHEMES

Scheme	Storage Cost			Communication	Execution Cost			
	User	Drone	Server	Cost	User	Drone	Server	Total
[22]	480	320	704	1472	$10T_h$	$7T_h$	$7T_h$	3175.01
[19]	808	512	768	1376	$6T_h + 2T_{sym} + T_{fe}$	$3T_h + 2T_{sym}$	$2T_h + 2T_{sym}$	35090.85
[25]	640	480	960	2336	$9T_h + 2T_{eccm}$	$3T_h + 2T_{eccm}$	$2T_h$	13750199.74
[35]	352	176	656	2208	$9T_h + T_{fe} + T_{eccm}$	$6T_h$	$7T_h + T_{eccm}$	3858247.09
[36]	1120	320	320	2061	$15T_h + T_{fe}$	$7T_h$	$10T_h + 2T_{sym}$	9556.10
[13]	512	352	672	2688	$8T_h$	$6T_h + 2T_{puf}$	$10T_h$	2706.39
[26]	640	320	384	1472	$14T_h$	$7T_h$	$7T_h$	3898.87
[8]	640	320	384	1472	$13T_h$	$5T_h$	$9T_h$	3384.16
[32]	1088	544	128	11585	$7T_h + T_{genR} + T_{encR} + T_{decR}$	$4T_h + T_{genR} + T_{encR}$	$6T_h + T_{decR}$	4468.64
Ours (R)	**1120**	**544**	**192**	**14530**	$\mathbf{6T_h + T_{fe} + T_{genR} + T_{encR} + T_{decR}}$	$\mathbf{4T_h + T_{genR} + T_{encR}}$	$\mathbf{6T_h + T_{decR}}$	**5721.68**
Ours (M)	**1120**	**544**	**192**	**33728**	$\mathbf{6T_h + T_{fe} + T_{genM} + T_{encM} + T_{decM}}$	$\mathbf{4T_h + T_{genM} + T_{encM}}$	$\mathbf{6T_h + T_{decM}}$	**7390.35**

function in computing PID, to mitigate such costs and facilitate comparison with other schemes.

Storage Cost: In our proposed protocol, the user, drone, and server store the values of $\{ss_i, fp_u, V_u, T_u, PID_u\}$, $\{ss_d, PID_d\}$, and $\{ID_u, PID_u, T_u, ID_d, PID_d, T_d\}$ with the total sizes of 1120 bits ($= 512 + 32 + 512 + 32 + 32$), 544 bits ($= 512 + 32$), and 192 bits ($= 32 + 32 + 32 + 32 + 32 + 32$), respectively.

Communication Cost: Three messages are transmitted during the authentication and session establishment process: $\mathtt{M}_1 = [\alpha_u, pk_u, h_u, T, PID_u]$ from U_i to CS with 13120 bits ($= 512 + 6400 + 6144 + 32 + 32$), $\mathtt{M}_2 = [\alpha_s, \beta_d, pk_u, T, PID_u]$ from CS to D_j with 7488 bits ($= 512 + 512 + 6400 + 32 + 32$), and $\mathtt{M}_3 = [\alpha_d, pk_d, h_d, T', PID_d]$ from D_j to U_i with 13120 bits ($= 512 + 6400 + 6144 + 32 + 32$).

Execution Cost: Table II outlines the notations and execution costs of various operations in nanoseconds (ns), estimated through implementation in C/C++, using the LatticeCrypto [38], MIRACL [39], and Liboqs [40] libraries [32], [41], [42]. We exclude the costs of bit-wise XOR, pseudo-random number generation, and concatenation. The test environment consists of a Windows 10 PC with 3.4 GHz Intel(R) i7-6700 CPU and 8 GB RAM as the server, and a Samsung GT-I9300 phone with Android 4.3, 1.4 GHz Quad-core Samsung Exynos 4412 processor, and 1 GB RAM as the user and drone devices. In our protocol, the user performs 6 hash operations, 1 fuzzy extractor, 1 key generation, 1 encapsulation, and 1 decapsulation, costing a total of 4551.71 ns ($= 6T_h + T_{fe} + T_{genM} + T_{encM} + T_{decM}$). The drone executes 4 hash operations, 1 key generation, and 1 encapsulation, amounting to 2753.58 ns ($= 4T_h + T_{genM} + T_{encM}$). The server utilizes 6 hash functions and 1 decapsulation, adding up to 85.05 ns ($= 6T_h + T_{decM}$). The overall execution cost of our MLWE-based scheme sums up to 7390.35 ns. If our proposed protocol adopts RLWE-based cryptography per [32]'s approach, with 4096 bits pk and 1 b h sizes: the storage costs remain unaffected, but our communication and execution costs would be 14530 bits ($= 4673 + 5184 + 4673$) and 5721.68 ns ($= 3717.035 + 1919.802 + 84.847$), respectively.

Table III compares the storage, communication, and execution costs of relevant authentication schemes. (R) and (M) denote the RLWE-based and MLWE-based results of our protocol, respectively. To contextualize our comparison, we briefly summarize the benchmark schemes: [22] uses hash functions but is vulnerable; [19] employs AEGIS and bio-metrics; ECC-based designs by [25], [35], and [36] emphasize on efficiency; [13] employs PUFs to avoid key storage, but none offer quantum-security; [26] applies RSA with fuzzy verification but was proven insecure by [8], who instead uses dynamic identities; and [32] proposed an RLWE-based scheme with notable vulnerabilities and limitations.

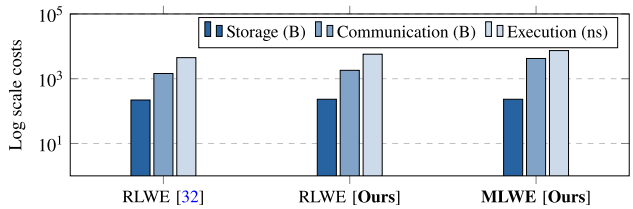


Fig. 6. Cost comparison of RLWE and MLWE-based approaches.

Our protocol has lower execution costs than [19], [25], [35], [36], and is comparable to other solutions while offering enhanced security and versatility. It addresses the design and security flaws of prior schemes, resists a broader range of attacks (Table I) and maintains acceptable performance. The increased overhead in lattice-based solutions stems from larger variables and stronger hashes required for quantum resistance. As shown in Fig. 6, the comparisons of RLWE and MLWE results indicate that these advantages are achieved with a justifiable overhead. Beyond aligning with PQC standards and enhancing security, MLWE offers flexibility across security levels and device capabilities, making our scheme suitable for diverse applications. The energy cost for communication is proportional to exchanged bits [14], [15], with negligible storage and computation overheads [14]. UAV energy profiles vary widely [4]; but since energy scales near-linearly with message size, our performance evaluation provides a practical proxy, as done in most comparable literature. By minimizing on-air traffic and optimizing functionalities, our design remains lightweight and energy-conscious. Although evaluated in a controlled setup, our protocol scales well, as each drone/user engages in a single session, while the server handles multiple connections with minimal costs. Given the server's assumed resource availability, the scheme remains practical even at large in large-scale networks.

VIII. CONCLUSION

In this paper, we proposed a comprehensive security scheme for drone networks; addressing challenges and vulnerabilities as well as the threats posed by quantum computers. To mitigate such threats, the proposed AKE scheme leverages PQC methods based on MLWE, error reconciliation, and hash functions. Our scheme employs ephemeral dynamic credentials, timestamps, multi-factor authentication, and context information to enhance

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.

communication security. The protocol also incorporates extensive procedures for node mobility, and dynamic node addition and revocation. The scheme's security was analyzed and proven through a sequence of games under the DY and BPR models and formally verified using the AVISPA tool. Our approach ensures confidentiality, integrity, and availability by providing mutual authentication, secure key establishment, forward security, and privacy; protecting the drone networks against a wide range of attacks. In addition to addressing the shortcomings of previous works, our evaluations confirm the scheme's robustness and effectiveness, showcasing a better trade-off between security, functionality, and performance. By enhancing drone communication security, our scheme paves the way for reliable and robust drone operations. Future work should focus on performance optimization and formal proofs under QROM.

REFERENCES

[1] E. T. Michailidis and D. Vouyioukas, "A review on software-based and hardware-based authentication mechanisms for the Internet of Drones," *Drones*, vol. 6, no. 2, 2022, Art. no. 41.

[2] Z. Wang et al., "A survey on cybersecurity attacks and defenses for unmanned aerial systems," *J. Syst. Archit.*, vol. 138, 2023, Art. no. 102870.

[3] M. Wazid, A. K. Das, N. Kumar, A. V. Vasilakos, and J. J. Rodrigues, "Design and analysis of secure lightweight remote user authentication and key agreement scheme in Internet of Drones deployment," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 3572–3584, Apr. 2019.

[4] M. Yahuza et al., "Internet of drones security and privacy issues: Taxonomy and open challenges," *IEEE Access*, vol. 9, pp. 57243–57 270, 2021.

[5] J.-P. Yaacoub, H. Noura, O. Salman, and A. Chehab, "Security analysis of drones systems: Attacks, limitations, and recommendations," *Internet Things*, vol. 11, 2020, Art. no. 100218.

[6] U. C. Cabuk, G. Dalkilic, and O. Dagdeviren, "CoMAD: Context-aware mutual authentication protocol for drone networks," *IEEE Access*, vol. 9, pp. 78400–78414, 2021.

[7] J. Sharma and P. S. Mehra, "Secure communication in IoT-based UAV networks: A systematic survey," *Internet Things*, vol. 23, 2023, Art. no. 100883.

[8] T.-F. Lee, D.-C. Lou, and C.-H. Chang, "Enhancing lightweight authenticated key agreement with privacy protection using dynamic identities for Internet of Drones," *Internet Things*, vol. 23, 2023, Art. no. 100877.

[9] N. I. of Standards and Technology, "Module-lattice-based key-encapsulation mechanism standard," U.S. Department of Commerce, StandardNIST FIPS 203 ipd, 2023.

[10] M. Bellare, D. Pointcheval, and P. Rogaway, "Authenticated key exchange secure against dictionary attacks," in *Proc. 19th Int. Conf. Theory Appl. Cryptographic Techn.*, 2000, pp. 139–155.

[11] D. Dolev and A. Yao, "On the security of public key protocols," *IEEE Trans. Inf. theory*, vol. 29, no. 2, pp. 198–208, Mar. 1983.

[12] A. Armando et al., "The AVISPA tool for the automated validation of internet security protocols and applications," in *Proc. Comput. Aided Verification: 17th Int. Conf.*, 2005, pp. 281–285.

[13] M. Sharma, B. Narwal, R. Anand, A. K. Mohapatra, and R. Yadav, "PSECAS: A physical unclonable function based secure authentication scheme for Internet of Drones," *Comput. Elect. Eng.*, vol. 108, 2023, Art. no. 108662.

[14] M. A. El-Zawawy, A. Brighente, and M. Conti, "SETCAP: Service-based energy-efficient temporal credential authentication protocol for Internet of Drones.," *Comput. Netw.*, vol. 206, pp. 108804–108814, 2022.

[15] "A lightweight and anonymous application-aware authentication and key agreement protocol for the Internet of Drones,"

[16] B. Bera, A. K. Das, and A. K. Sutrala, "Private blockchain-based access control mechanism for unauthorized UAV detection and mitigation in Internet of Drones environment," *Comput. Commun.*, vol. 166, pp. 91–109, 2021.

[17] A. Yazdinejad, R. M. Parizi, A. Dehghantanha, and H. Karimipour, "Federated learning for drone authentication," *Ad Hoc Netw.*, vol. 120, 2021, Art. no. 102574.

[18] C. S. Veerappan, P. K. K. Loh, and R. J. Chennattu, *Smart Drone Controller Framework-Toward an Internet of Drones*. Berlin, Germany: Springer, 2022, pp. 1–14.

[19] M. Tanveer, A. H. Zahid, M. Ahmad, A. Baz, and H. Alhakami, "LAKE-IoD: Lightweight authenticated key exchange protocol for the Internet of Drone environment," *IEEE Access*, vol. 8, pp. 155645–155659, 2020.

[20] J. Li, Y. Wang, Y. Ding, W. Wu, C. Li, and H. Wang, "A certificateless pairing-free authentication scheme for unmanned aerial vehicle networks," *Secur. Commun. Netw.*, vol. 2021, 2021, Art. no. 9463606.

[21] G. Cho, J. Cho, S. Hyun, and H. Kim, "SENTINEL: A secure and efficient authentication framework for unmanned aerial vehicles," *Appl. Sci.*, vol. 10, no. 9, 2020, Art. no. 3149.

[22] Y. Zhang, D. He, L. Li, and B. Chen, "A lightweight authentication and key agreement scheme for Internet of Drones," *Comput. Commun.*, vol. 154, pp. 455–464, 2020.

[23] S. U. Jan, I. A. Abbasi, and F. Algarni, "A key agreement scheme for IoD deployment civilian drone," *IEEE Access*, vol. 9, pp. 149311–149321, 2021.

[24] C. F. E. de Melo et al., "Uavouch: A secure identity and location validation scheme for UAV-networks," *IEEE Access*, vol. 9, pp. 82930–82946, 2021.

[25] M. Nikooghadam, H. Amintoosi, S. H. Islam, and M. F. Moghadam, "A provably secure and lightweight authentication scheme for Internet of Drones for smart city surveillance," *J. Syst. Archit.*, vol. 115, 2021, Art. no. 101955.

[26] D. Chaudhary, T. Soni, K. L. Vasudev, and K. Saleem, "A modified lightweight authenticated key agreement protocol for Internet of Drones," *Internet Things*, vol. 21, 2023, Art. no. 100669.

[27] J. Ding, X. Xie, and X. Lin, "A simple provably secure key exchange scheme based on the learning with errors problem," *Cryptol. ePrint Arch.*, 2012, Paper 2012/688. [Online]. Available: https://eprint.iacr.org/2012/688

[28] P. Ren, X. Gu, and Z. Wang, "Efficient module learning with errors-based post-quantum password-authenticated key exchange," *IET Inf. Secur.*, vol. 17, no. 1, pp. 3–17, 2023.

[29] C. Peikert, "Lattice cryptography for the Internet," in *Proc. Int. Workshop Post-Quantum Cryptogr.*, 2014, pp. 197–219.

[30] J. Bos et al., "FRODO: Take off the ring! practical, quantum-secure key exchange from LWE," in *Proc. 2016 ACM SIGSAC Conf. Comput. Commun. Secur.*, 2016, pp. 1006–1018.

[31] S. Guo, Y. Song, S. Guo, Y. Yang, and S. Song, "Three-party password authentication and key exchange protocol based on MLWE," *Symmetry*, vol. 15, no. 9, 2023, Art. no. 1750.

[32] D. Mishra et al., "Quantum-safe secure and authorized communication protocol for Internet of Drones," *IEEE Trans. Veh. Technol.*, vol. 72, no. 12, pp. 16499–16 507, Dec. 2023.

[33] J. Ding, S. Alsayigh, R. Saraswathy, S. Fluhrer, and X. Lin, "Leakage of signal function with reused keys in RLWE key exchange," in *Proc. IEEE Int. Conf. Commun.*, 2017, pp. 1–6.

[34] V. Shoup, "Sequences of games: A tool for taming complexity in security proofs," *Cryptology ePrint Arch.*, 2004, Paper 2004/332. [Online]. Available: https://eprint.iacr.org/2004/332

[35] S. Hussain, S. A. Chaudhry, O. A. Alomari, M. H. Alsharif, M. K. Khan, and N. Kumar, "Amassing the security: An ECC-based authentication scheme for Internet of Drones," *IEEE Syst. J.*, vol. 15, no. 3, pp. 4431–4438, Sep. 2021.

[36] S. Hussain, K. Mahmood, M. K. Khan, C.-M. Chen, B. A. Alzahrani, and S. A. Chaudhry, "Designing secure and lightweight user access to drone for smart city surveillance," *Comput. Standards Interfaces*, vol. 80, 2022, Art. no. 103566.

[37] National Institute of Standards and Technology, "SHA-3 standard: Permutation-based hash and extendable-output functions," U.S. Department of Commerce, Washington, D.C., Standard NIST FIPS 202 ipd, 2015, doi: 10.6028/NIST.FIPS.202.

[38] P. Longa and M. Naehrig, "Speeding up the number theoretic transform for faster ideal lattice-based cryptography," in *Proc. Cryptol. Netw. Secur.: 15th Int. Conf.*, 2016, pp. 124–139.

[39] M. Community, "Miracl crypto library," [Online]. Available: 2018. [Online]. Available: https://github.com/miracl/MIRACL

[40] D. Stebila and M. Mosca, "Post-quantum key exchange for the Internet and the open quantum safe project," in *Proc. Int. Conf. Sel. Areas Cryptogr.*, 2016, pp. 14–37.

[41] J. Zhang, Z. Zhang, J. Ding, M. Snook, and Ö. Dagdelen, "Authenticated key exchange from ideal lattices," in *Proc. 34th Int. Conf. Theory Appl. Cryptographic Techn.*, 2015, pp. 719–751.

[42] Q. Feng, D. He, S. Zeadally, N. Kumar, and K. Liang, "Ideal lattice-based anonymous authentication protocol for mobile devices," *IEEE Syst. J.*, vol. 13, no. 3, pp. 2775–2785, Sep. 2019.

Authorized licensed use limited to: Mepco Schlenk Engineering College. Downloaded on July 27,2026 at 05:05:28 UTC from IEEE Xplore. Restrictions apply.